

WISCONSIN UNIVERSITY

Management System — Django Models Reference

This document lists all database models (tables) used in the Django project. Each model is explained in simple English so anyone on the team can understand what data is stored and why.

Total Apps	Total Models	Total Fields	Database
17	49	362+	PostgreSQL

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1. ACCOUNTS — User Login & Roles

This handles who can log in to the system and what they are allowed to do.

Model: User

Every person who uses the system — student, teacher, admin — has one User account with a username and password.

Field Name	What it stores	Notes
username	The login name used to sign in	Must be unique
email	Email address of the person	Must be unique
role	What type of user: admin, student, staff, faculty, parent, librarian, wa	Required
phone	Phone number	Optional
profile_photo	A photo of the person	Optional
is_active	Is this account turned on or off?	Yes/No
created_at	When the account was created	Auto-filled
updated_at	When the account was last changed	Auto-filled

2. STUDENTS — Student Profiles

Stores all information about students — their personal details, family contacts, and uploaded documents.

Model: Student

The main profile of a student. Every student has one profile with their personal and contact information.

Field Name	What it stores	Notes
student_id	Unique ID for the student e.g. WIS-2024-00123	Auto-generated
ssn	Social Security Number or National ID	Must be unique
first_name	Student's first name	Required
last_name	Student's last name	Required
date_of_birth	Date the student was born	Required
gender	Male, Female, or Other	Required
blood_group	Blood type e.g. A+, O-	Optional
nationality	Country they are from	Default: American
address	Home address	Required
city	City	Required
state	State	Required
zip_code	ZIP/Postal code	Required
email	Personal email address	Must be unique
phone	Mobile phone number	Required
emergency_phone	Phone number to call in emergencies	Optional
status	Is the student active, graduated, suspended etc.	Required
admission_date	The date they joined the university	Required
photo	Student's passport-size photo	Optional

Model: Guardian

Parent or guardian details for each student. One student can have more than one guardian listed.

Field Name	What it stores	Notes
student	Which student this guardian belongs to	Required
name	Full name of the parent/guardian	Required
relation	Father, Mother, Guardian, or Sibling	Required
phone	Their phone number	Required
email	Their email address	Optional
occupation	Their job	Optional
annual_income	How much they earn per year	Optional
is_primary	Is this the main contact person?	Yes/No

Model: StudentDocument

Documents uploaded for a student like birth certificate, ID proof, etc.

Field Name	What it stores	Notes
student	Which student this document belongs to	Required
doc_type	Type: Birth Cert, ID Proof, Transcript, Photo, Medical, Other	Required
title	Name or description of the document	Required
file	The actual uploaded file	Required
uploaded_at	When it was uploaded	Auto-filled
verified	Has an admin checked and approved this?	Yes/No

3. ADMISSIONS — Applications

Manages the process when new students apply to join the university.

Model: AdmissionCycle

One round of admissions e.g. "Fall 2024". All applications belong to a cycle.

Field Name	What it stores	Notes
name	Name of the cycle e.g. Fall 2024	Required
year	The year	Required
start_date	When applications open	Required
end_date	When applications close	Required
is_active	Is this cycle currently open?	Yes/No

Model: Application

One application form submitted by a person who wants to join the university.

Field Name	What it stores	Notes
application_no	Unique application number (auto-generated)	Auto
cycle	Which admission round this belongs to	Required
applicant_name	Full name of the applicant	Required
email	Their email	Required
phone	Their phone number	Required
date_of_birth	Date of birth	Required
address	Current address	Required

Field Name	What it stores	Notes
program_applied	Which course/program they applied for	Required
previous_school	Name of their previous school/college	Optional
previous_gpa	Their GPA from previous studies	Optional
status	Where it is: Draft, Submitted, Under Review, Accepted, Rejected etc	Required
remarks	Notes added by the admin	Optional
submitted_at	When the form was submitted	Optional

4. STAFF — Teachers & Employees

Stores information about all staff members including teachers, office staff, and their departments.

Model: Department

A department in the university e.g. Computer Science, English, Administration.

Field Name	What it stores	Notes
name	Full name of the department	Required
code	Short code e.g. CS, ENG	Must be unique
description	About this department	Optional
head	Who is in charge of this department	Optional

Model: Designation

Job titles used in the university e.g. Professor, Lecturer, Clerk.

Field Name	What it stores	Notes
name	Job title name	Required
department	Which department this title belongs to	Optional
grade	Pay grade or band	Optional

Model: Staff

Full profile of a staff or faculty member including job details and salary.

Field Name	What it stores	Notes
staff_id	Unique staff ID number	Auto
first_name	First name	Required
last_name	Last name	Required
ssn	Social Security Number	Must be unique
department	Which department they work in	Required
designation	Their job title	Required
employment_type	Full Time, Part Time, Contract, or Visiting	Required
date_of_joining	When they started working here	Required
date_of_birth	Date of birth	Required
gender	Male, Female, or Other	Required
phone	Phone number	Required
email	Official email address	Must be unique
qualification	Highest degree or qualification	Optional

Field Name	What it stores	Notes
experience_years	How many years of experience they have	Required
salary	Monthly salary amount	Required
bank_account	Bank account number for salary transfer	Optional
is_active	Are they currently employed?	Yes/No

5. COURSES — Programs & Subjects

Manages everything about courses — what programs are offered, what subjects exist, and which students are enrolled.

Model: AcademicYear

One full academic year e.g. 2024-2025. All records like results and fees are tied to a year.

Field Name	What it stores	Notes
name	e.g. 2024-2025	Required
start_date	When the year starts	Required
end_date	When the year ends	Required
is_current	Is this the current running year?	Yes/No

Model: Program

A degree program offered by the university e.g. B.Sc. Computer Science, MBA.

Field Name	What it stores	Notes
name	Full program name	Required
code	Short code e.g. BSCS	Must be unique
department	Which department offers this	Required
level	Certificate, Diploma, Bachelors, Masters, PhD	Required
duration	How many semesters it takes to complete	Required
total_credits	Total credits needed to graduate	Default 120
is_active	Is this program currently running?	Yes/No

Model: Course

One subject or class inside a program e.g. Mathematics 101, Introduction to Python.

Field Name	What it stores	Notes
code	Subject code e.g. CS101	Must be unique
name	Subject name	Required
program	Which program this subject belongs to	Required
faculty	Which teacher teaches this	Optional
credits	Number of credit hours	Default 3
course_type	Core, Elective, Lab, or Seminar	Required
semester	Which semester number e.g. 1, 2, 3	Required
description	What this subject is about	Optional
max_students	Maximum number of students allowed	Default 60
is_active	Is this subject running this year?	Yes/No

Model: Enrollment

When a student signs up for a subject, this record is created.

Field Name	What it stores	Notes
student	Which student	Required
course	Which subject they enrolled in	Required
academic_year	Which year	Required
status	Enrolled, Dropped, or Completed	Required
enrolled_on	Date of enrollment	Auto-filled

6. TIMETABLE — Class Schedule

Manages classrooms and the weekly timetable for all subjects.

Model: Room

A physical room in the university — classrooms, labs, or halls.

Field Name	What it stores	Notes
number	Room number e.g. 101, Lab-3	Must be unique
building	Which building it is in	Required
floor	Floor number (0 = ground floor)	Required
capacity	How many students can sit inside	Required
room_type	Classroom, Lab, Hall, or Seminar Room	Required
has_projector	Does it have a projector or smart board?	Yes/No
is_active	Is this room available to use?	Yes/No

Model: TimetableEntry

One class in the weekly timetable — which subject, which room, which teacher, on which day and time.

Field Name	What it stores	Notes
course	Which subject	Required
room	Which room the class is in	Required
faculty	Which teacher is taking the class	Required
academic_year	Which year this timetable is for	Required
day	Day of week: Mon, Tue, Wed, Thu, Fri, Sat	Required
start_time	What time the class starts	Required
end_time	What time the class ends	Required

7. EXAMS — Tests & Results

Everything about exams — scheduling them, recording student marks, and issuing transcripts.

Model: Exam

One exam event — e.g. the Midterm for Mathematics 101 on 15 Oct 2024.

Field Name	What it stores	Notes
name	Name of the exam e.g. Midterm Exam	Required
course	Which subject this exam is for	Required

Field Name	What it stores	Notes
exam_type	Midterm, Final, Quiz, Assignment, or Practical	Required
exam_date	Date of the exam	Required
start_time	What time it starts	Required
end_time	What time it ends	Required
room	Where the exam is held	Optional
total_marks	Maximum marks for this exam	Default 100
pass_marks	Minimum marks needed to pass	Default 40
is_published	Have the results been released to students?	Yes/No

Model: ExamResult

The marks scored by one student in one exam.

Field Name	What it stores	Notes
student	Which student	Required
exam	Which exam	Required
marks_obtained	How many marks they scored	Required
grade	Letter grade: A+, A, B+, B, C, D, F	Required
is_absent	Was the student absent for this exam?	Yes/No
remarks	Any extra notes from the teacher	Optional
graded_by	Which teacher gave this grade	Optional

Model: Transcript

The official marks sheet for a student for one full academic year.

Field Name	What it stores	Notes
student	Which student	Required
academic_year	Which year this transcript covers	Required
gpa	Overall GPA for that year	Required
total_credits	Total credits earned that year	Required
issued_date	Date the transcript was issued	Auto-filled
issued_by	Staff member who issued it	Optional
file	The PDF transcript file	Optional

8. ATTENDANCE — Daily Presence

Tracks whether students came to class each day, and manages leave requests.

Model: Attendance

One attendance record for one student in one class on one day.

Field Name	What it stores	Notes
student	Which student	Required
course	Which class/subject	Required
date	The date	Required
status	Present, Absent, Late, or Excused	Required

Field Name	What it stores	Notes
marked_by	Which teacher marked it	Optional
remarks	Any notes e.g. reason for absence	Optional
marked_at	When attendance was recorded	Auto-filled

Model: LeaveApplication

When a student wants to take leave, they submit this form.

Field Name	What it stores	Notes
student	Which student is applying	Required
leave_type	Medical, Personal, Family Emergency, or Other	Required
from_date	Leave starts from this date	Required
to_date	Leave ends on this date	Required
reason	Why they need leave	Required
document	Supporting proof e.g. doctor certificate	Optional
status	Pending, Approved, or Rejected	Required
approved_by	Which staff approved or rejected it	Optional
applied_at	When the request was submitted	Auto-filled

9. FINANCE — Fees & Payments

Handles all money matters — fee amounts, bills sent to students, payments received, and scholarships.

Model: FeeStructure

The fee plan — how much a student in a specific program needs to pay and for what.

Field Name	What it stores	Notes
program	Which program this fee is for	Required
academic_year	Which year	Required
fee_type	Tuition, Hostel, Transport, Library, Sports, Exam, Lab, or Misc	Required
amount	How much in dollars	Required
due_date	When this fee must be paid by	Required
is_optional	Is this fee compulsory or optional?	Yes/No

Model: Invoice

The bill sent to a student showing how much they owe.

Field Name	What it stores	Notes
invoice_no	Unique bill number	Must be unique
student	Which student this bill is for	Required
academic_year	Which year	Required
total_amount	Total amount owed	Required
amount_paid	How much has been paid so far	Default 0
status	Pending, Partially Paid, Paid, Overdue, or Waived	Required
due_date	Last date to pay	Required
issued_at	When the bill was created	Auto-filled

Model: Payment

One payment made by a student towards their bill.

Field Name	What it stores	Notes
invoice	Which bill this payment is for	Required
amount	How much was paid	Required
mode	Cash, Card, Bank Transfer, Online, or Cheque	Required
transaction_id	Reference number from bank or payment app	Optional
receipt_no	Receipt number given to student	Optional
paid_at	When payment was made	Auto-filled
collected_by	Staff who received the payment	Optional

Model: Scholarship

A scholarship or fee waiver given to a student.

Field Name	What it stores	Notes
student	Which student got the scholarship	Required
name	Name of the scholarship	Required
scholarship_type	Merit, Need Based, Sports, Government, or Other	Required
amount	Money amount of the scholarship	Required
academic_year	Which year it is for	Required
awarded_date	When it was given	Required
valid_till	When it expires	Optional

10. SPORTS — Sports Academy

Manages all sports activities — teams, players, competitions, and achievements.

Model: SportsCategory

The type of sport e.g. Basketball, Cricket, Swimming.

Field Name	What it stores	Notes
name	Sport name	Must be unique
is_indoor	Is it played indoors?	Yes/No
description	About the sport	Optional

Model: SportsTeam

A university team for a specific sport.

Field Name	What it stores	Notes
name	Team name	Required
category	Which sport	Required
gender	Male, Female, or Mixed team	Required
coach	Staff member who coaches this team	Optional
captain	Student who is the captain	Optional
established	Year the team was formed	Optional

Field Name	What it stores	Notes
achievements	List of notable wins or awards	Optional
is_active	Is this team currently active?	Yes/No

Model: Athlete

A student who is a registered player in a sports team.

Field Name	What it stores	Notes
student	Which student	Required
team	Which team they play in	Required
position	Player, Captain, Vice Captain, or Reserve	Required
jersey_no	Their jersey/kit number	Optional
joined_date	When they joined the team	Auto-filled
is_active	Are they currently in the team?	Yes/No

Model: SportsEvent

A sports match or tournament the university takes part in.

Field Name	What it stores	Notes
name	Event name	Required
category	Which sport	Required
level	Internal, Inter-University, State, National, or International	Required
venue	Where it is held	Required
start_date	Start date	Required
end_date	End date	Required
organizer	Who is organizing it	Optional
teams	Which university teams are participating	Many teams

Model: SportsAchievement

A prize or ranking won by a team or student in a sports event.

Field Name	What it stores	Notes
event	Which event	Required
team	Which team won	Optional
student	Which individual student won	Optional
position	1st (Gold), 2nd (Silver), 3rd (Bronze), or Participant	Required
award_name	Name of the award or trophy	Optional
certificate	Certificate file	Optional

11. LIBRARY — Books & Borrowing

Manages the university library — what books are available and who has borrowed them.

Model: BookCategory

Category or subject type of a book e.g. Science, History, Engineering.

Field Name	What it stores	Notes
name	Category name	Must be unique

Model: Book

One book in the library collection.

Field Name	What it stores	Notes
isbn	Unique book identification number	Must be unique
title	Name of the book	Required
author	Author name	Required
publisher	Publisher name	Optional
edition	Edition number	Optional
category	Which category the book belongs to	Optional
published_year	Year of publication	Optional
language	Language the book is in	Default: English
total_copies	Total number of copies owned	Default 1
available_copies	How many copies are currently available	Default 1
location	Where it is placed on the shelf	Optional

Model: BookIssue

A record of someone borrowing a book from the library.

Field Name	What it stores	Notes
book	Which book was borrowed	Required
issued_to	Who borrowed it (student or staff)	Required
issue_date	Date it was taken	Auto-filled
due_date	Date it must be returned by	Required
return_date	Date it was actually returned	Optional
status	Issued, Returned, Overdue, or Lost	Required
fine_amount	Fine charged if overdue or lost	Default 0
fine_paid	Has the fine been paid?	Yes/No

12. HOSTEL — Student Housing

Manages hostel buildings, individual rooms, and which student lives in which room.

Model: Hostel

A hostel or dormitory building on campus.

Field Name	What it stores	Notes
name	Name of the hostel	Required
gender	Male, Female, or Mixed	Required
address	Physical address of the building	Required
warden	Staff member in charge of this hostel	Optional
total_rooms	Total number of rooms in this hostel	Required
is_active	Is this hostel open?	Yes/No

Model: HostelRoom

One individual room inside a hostel.

Field Name	What it stores	Notes
hostel	Which hostel this room is in	Required
room_number	Room identifier e.g. 101, A-22	Required
floor	Floor level (0 = ground)	Required
room_type	Single, Double, Triple, or Dormitory	Required
capacity	Max number of students in this room	Default 2
current_occupancy	How many students are in it now	Default 0
monthly_rent	Monthly rent for this room	Required
has_ac	Does it have air conditioning?	Yes/No
is_available	Is this room free right now?	Yes/No

Model: HostelAllotment

When a student is assigned to a specific hostel room.

Field Name	What it stores	Notes
student	Which student	Required
room	Which room they are assigned to	Required
allotted_on	Date they moved in	Auto-filled
vacated_on	Date they moved out	Optional
status	Active, Vacated, or Transferred	Required
remarks	Any notes	Optional

13. TRANSPORT — Buses & Routes

Manages the university bus service — routes, vehicles, and which students use which bus.

Model: TransportRoute

One bus route with all its stops.

Field Name	What it stores	Notes
route_number	Route code e.g. R01	Must be unique
name	Route name e.g. North Campus Route	Required
start_point	Where the bus starts from	Required
end_point	Where the bus ends	Required
stops	List of all bus stops along the way	Required
distance_km	Total distance in kilometres	Optional
monthly_fee	Monthly fee for students using this route	Required
is_active	Is this route running?	Yes/No

Model: Vehicle

A bus or van owned by the university.

Field Name	What it stores	Notes
reg_number	Vehicle registration plate number	Must be unique
model	Make and model of the vehicle	Required
capacity	How many passengers it can carry	Required
driver	Who drives this vehicle	Optional
route	Which route this vehicle is on	Optional
insurance_expiry	When the insurance expires	Optional
is_active	Is this vehicle in service?	Yes/No

Model: TransportAllotment

When a student is registered to use a bus route.

Field Name	What it stores	Notes
student	Which student	Required
route	Which route they use	Required
pickup_stop	Which bus stop they board from	Required
allotted_on	Date they registered	Auto-filled
is_active	Are they currently using this route?	Yes/No

14. HR — Staff Leave & Salary

Human Resources — managing staff leave, monthly salaries, and yearly performance reviews.

Model: LeaveType

The different types of leave that staff can take e.g. Sick Leave, Annual Leave.

Field Name	What it stores	Notes
name	Name of the leave type	Must be unique
max_days	Maximum days allowed per year	Required
is_paid	Is salary paid during this leave?	Yes/No
is_carryover	Can unused days carry over to next year?	Yes/No

Model: StaffLeave

A leave application submitted by a staff member.

Field Name	What it stores	Notes
staff	Which staff member	Required
leave_type	What type of leave	Required
from_date	Leave starts from	Required
to_date	Leave ends on	Required
reason	Why they need leave	Required
status	Pending, Approved, Rejected, or Cancelled	Required
approved_by	Who approved or rejected it	Optional
applied_at	When the request was made	Auto-filled

Model: Payroll

Monthly salary record for one staff member.

Field Name	What it stores	Notes
staff	Which staff member	Required
month	Month number (1 to 12)	Required
year	Year of this payroll	Required
basic_salary	Basic monthly salary	Required
allowances	Extra allowances added	Default 0
deductions	Deductions like loans or penalties	Default 0
tax	Income tax deducted	Default 0
net_salary	Final take-home salary	Required
status	Draft, Processed, or Paid	Required
paid_on	Date salary was transferred	Optional

Model: Appraisal

Yearly performance review of a staff member by their manager.

Field Name	What it stores	Notes
staff	Which staff is being reviewed	Required
academic_year	Which year	Required
appraiser	Who is doing the review (manager/HOD)	Optional
rating	Rating from 1 (poor) to 5 (excellent)	Required
strengths	What they are doing well	Optional
improvements	What they need to improve	Optional
goals_next_year	Goals set for the coming year	Optional
remarks	Any extra notes	Optional

15. EVENTS — Clubs & Activities

Manages student clubs, club memberships, and university events like fests and ceremonies.

Model: Club

A student club or society e.g. Drama Club, Coding Club, Music Society.

Field Name	What it stores	Notes
name	Club name	Required
category	Type: Cultural, Technical, Social etc.	Optional
description	What the club is about	Optional
faculty_advisor	Teacher who advises this club	Optional
president	Student who leads the club	Optional
founded_year	Year the club was started	Optional
is_active	Is this club currently active?	Yes/No

Model: ClubMember

A student who is a member of a club.

Field Name	What it stores	Notes
club	Which club	Required
student	Which student	Required
role	Member, Secretary, Treasurer, President, Vice President	Required
joined_date	When they joined	Auto-filled
is_active	Are they still an active member?	Yes/No

Model: UniversityEvent

A university-level event like Annual Day, Tech Fest, Convocation.

Field Name	What it stores	Notes
name	Event name	Required
event_type	Cultural, Technical, Sports, Academic, Convocation, Other	Required
description	About the event	Optional
venue	Where it is held	Required
start_date	Start date and time	Required
end_date	End date and time	Required
organizer	Staff member organizing it	Optional
is_public	Is it open to the public?	Yes/No
registration_open	Can people register to attend?	Yes/No
max_participants	Maximum number of attendees	Optional

16. NOTIFICATIONS — Messages & Alerts

Handles announcements posted by admin and individual notifications sent to users.

Model: Announcement

A message posted by admin to everyone or to a specific group.

Field Name	What it stores	Notes
title	Title of the announcement	Required
content	Full message content	Required
audience	Who it is for: All, Students, Staff, or Parents	Required
is_urgent	Should this be shown as urgent/important?	Yes/No
attachment	Optional file attached to announcement	Optional
posted_by	Who posted it	Optional
published_at	When it was posted	Auto-filled
expires_at	When it should stop showing	Optional

Model: Notification

A personal notification sent to one specific user e.g. "Your fee is due".

Field Name	What it stores	Notes
user	Which user gets this notification	Required
title	Short title	Required
message	Full message	Required

Field Name	What it stores	Notes
n_type	Type: Info, Warning, Success, or Danger	Required
is_read	Has the user read it?	Yes/No
link	Optional link to click (URL)	Optional
created_at	When it was created	Auto-filled

17. CERTIFICATES — Certificates & Requests

Manages certificate types and student requests for official documents.

Model: Certificate

A type of certificate the university can issue e.g. Bonafide Certificate, Transfer Certificate.

Field Name	What it stores	Notes
name	Certificate name	Required
cert_type	Bonafide, Transfer, Conduct, Completion, Sports, Other	Required
template	Template file used to generate the certificate	Optional
description	What this certificate is for	Optional

Model: CertificateRequest

When a student asks for a certificate to be issued to them.

Field Name	What it stores	Notes
student	Which student is requesting	Required
certificate	Which type of certificate they want	Required
purpose	Why they need it e.g. for visa application	Required
status	Pending, Processing, Ready, Issued, or Rejected	Required
requested_at	When the request was made	Auto-filled
issued_at	When it was given to the student	Optional
issued_file	The generated certificate PDF	Optional
processed_by	Staff who handled the request	Optional
remarks	Notes or reason if rejected	Optional

How to Set Up This Project

1. Install Django and create project

```
pip install django psycopg2-binary pillow
django-admin startproject wisconsin_project
cd wisconsin_project
```

2. Create all 17 apps

```
python manage.py startapp accounts
python manage.py startapp students
python manage.py startapp admissions
... (do this for all 17 apps)
```

3. Set the custom User model in settings.py

```
AUTH_USER_MODEL = 'accounts.User'
```

4. Add all apps to INSTALLED_APPS in settings.py

```
INSTALLED_APPS = [
    ...
    'accounts', 'students', 'admissions', 'staff',
    'courses', 'timetable', 'exams', 'attendance',
    'finance', 'sports', 'library', 'hostel',
    'transport', 'hr', 'events', 'notifications', 'certificates',
]
```

5. Copy the model code into each app's models.py

Take the model classes from models_all.py and paste them into the correct app folder.

6. Run database migrations

```
python manage.py makemigrations
python manage.py migrate
```

7. Create an admin account and start the server

```
python manage.py createsuperuser
python manage.py runserver
```

8. Open the admin panel

Go to <http://127.0.0.1:8000/admin> in your browser
Login with the superuser account you created